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Dual-view collaboration fusion on diffusion learning for bundle recommendation

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ABSTRACT

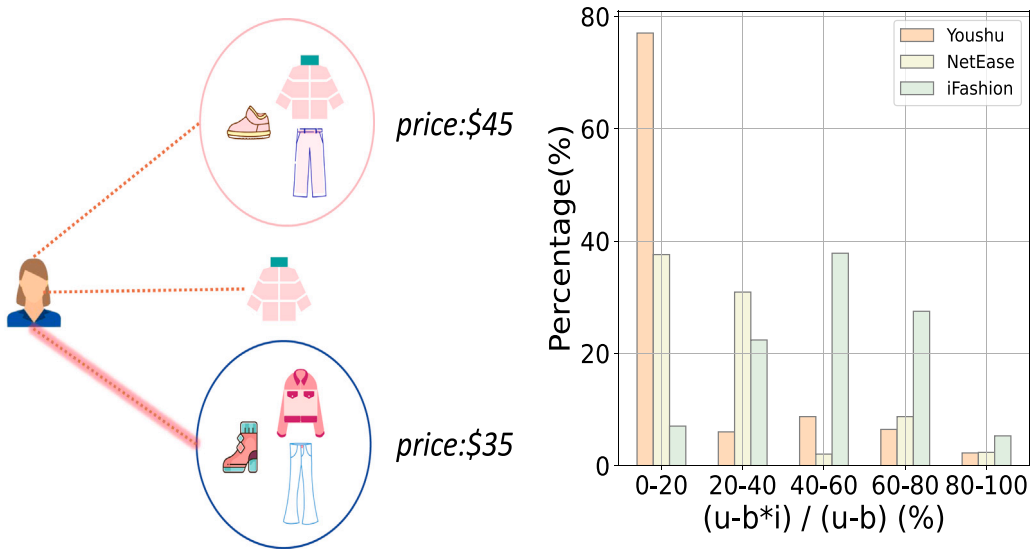
Bundle recommendation is a popular sales strategy in the e-commerce domain, centered around offering combinations of related items in a single order. Unlike traditional recommendation approaches that prioritize individual items, bundle recommendation focuses on recommending groups of items. Recent methods typically model three interaction types (user–bundle, user–item, and bundle–item) using separate views based on processes related to graph neural networks. However, these methods primarily emphasize integrating information from different views, often neglecting the negative impact of item-level information on the diversity of bundle recommendations. To tackle the problem, we propose the information fusion-based **D**ual-view **C**ollaboration on **D**iffusion **L**earning (DCDL) framework for bundle recommendation. It employs a dual-view structure, applying targeted methods based on the characteristics of the views, and integrates a diffusion learning paradigm tailored for user features. This paradigm optimally adjusts the feature distribution more uniformly by leveraging intermediate outputs in the denoising phase to enhance recommendation diversity. Experiments conducted on three public datasets validate the effectiveness of DCDL.

1. Introduction

Unlike traditional item recommendation strategies [1,2], bundle recommendation strategies have gained significant popularity. On the one hand, they meet the potential needs of users through the assembly of multiple items. On the other hand, they support users in purchasing multiple items at once, thereby avoiding redundant orders and transportation costs. A bundle typically refers to a group of complementary items that are related to a specific life scenario, such as a bundle that includes a computer, mouse, and headphones. Another approach to bundling involves grouping items within the same thematic category, such as a music collection themed around "sadness" or novels centered on "dreams". These different bundling strategies aim to comprehensively meet users' real-life or interest-based needs in a convenient, one-stop manner. Users only need to order a single bundle to receive all the items contained within it. By ordering a single bundle, users receive all the included items, often at a lower cost than purchasing each item individually. Since bundling facilitates the transportation of multiple items together, shipping costs are reduced, allowing the bundle price to be lower than the sum of its parts. Consumer psychology [3] suggests that people are inclined to purchase all the items they need or desire at a particular stage of a transaction. Bundle recommendation strategies align with this phenomenon and can simplify consumers' purchasing decisions. Moreover, during the process of constructing bundles, it is common to combine popular items with less popular ones, which naturally helps alleviate the long-tail distribution problem in recommendation systems to some extent. This scenario includes three fundamental entities: users, bundles, and items, along with three types of interactive relationships: user–bundle (U–B), bundle–item (B–I), and user–item (U–I). Enhancing the predictive accuracy for potential U–B interactions is the objective of the bundle recommendation task.

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(a) Diversity of user choices for bundles.

(b) Proportion of users by the γ value.

Fig. 1. 'User diversity choices' case and data validation.

Early strategies for addressing the bundle recommendation task utilized matrix factorization and multitask learning frameworks. For instance, MFBPR [4] employed the matrix factorization and Bayesian personalized ranking (BPR) loss calculation to construct and optimize the model. DAM [5] utilized a multitask learning framework to jointly enhance sequential recommendation and bundle recommendation tasks. With the rise of Graph Neural Networks [6] in the information retrieval domain, more sophisticated methods have been developed. BGCN [7] was the first to model the task from two views, capturing more semantics through the integration of the two views. MIDGN [8] attempted to disentangle multiple intents potentially present in a bundle under global and local views with a contrastive learning framework. CrossCBR [9] simplified the information propagation method under two views using LightGCN [10] and added contrastive learning constraints between views. However, users inherently exhibit diversity in their choices of bundles, and unconstrained integration of item information can undermine this diversity, despite the bundles being composed of items. For instance, as illustrated in Fig. 1(a), users may choose bundles that do not include items they have previously interacted with due to factors such as cost-effectiveness(colored interaction line represents diversity interaction). Besides, for three real-world datasets, YouShu [5] (books as items and booklists as bundles), NetEase [11] (single music tracks as items and playlists as bundles), and iFashion [12] (clothing pieces as items and outfits as bundles), we calculated the γ, representing the ratio of the number of bundles a user interacted with that contain the items the user has interacted with (u-b*i) to the total number of bundles the user has interacted with (u–b). Based on the γ value for each user, we categorized users into different statistical intervals as shown in Fig. 1(b). We can see that in these datasets, a large amount of U–B interactions do not rely on item-level information, with Youshu showing this trend most prominently. This data-driven evidence demonstrates that, although users might intuitively prefer bundles containing their favorite items, their actual choices are far more diverse. This highlights two key challenges in the field:

- How can dual views be specifically modeled and efficiently learned in a bundle recommendation scenario?
- How can diversity in bundle recommendations be effectively maintained during the collaboration of different views?

To address the above challenges, we propose a novel information fusion-based Dual-view Collaboration on Diffusion Learning (DCDL) framework to address the bundle recommendation task. First, we reference prior work's [9] dual-view partitioning scheme (bundle view and item view), and propose tailored processing methodologies designed for the unique data characteristics of each view. Second, we apply the diffusion learning training paradigm to optimize the feature distribution after view information fusion, ensuring a more uniform distribution and thereby enhancing recommendation diversity. To the best of our knowledge, this is the first instance of applying diffusion learning to the bundle recommendation field.

Specifically, to tackle the sparsity of U–B interactions, we employ a graph attention network (GAT, [13]) to process the bipartite graph, further enhancing relationships with an augmented relation matrix. For the large volume of U–I interactions, we capture the similarity between items based on their co-occurrence relationships to update item features and aggregate them to obtain the corresponding user/bundle's feature. Additionally, we innovatively design a collaborative fusion diffuser(CFD), which treats the original user features under the bundle view as the initial state, adds noise to them, and uses the fused user features as auxiliary information in the denoising phase to obtain an intermediate product. This intermediate product is then used to guide feature distribution. Compared to the existing methods, DCDL not only provides specific modeling for dual views but also addresses the challenge of recommendation diversity by optimizing feature distribution.

Our contributions can be summarized as follows:

- We develop tailored methods to model each view internally, taking into account the unique characteristics of its data.
- We propose the DCDL framework, which incorporates a diffusion learning module to optimize feature distribution and enhance collaboration between views, thereby improving diversity in bundle recommendations.
- We conduct extensive experiments on three public datasets, demonstrating the effectiveness of the DCDL framework in addressing the bundle recommendation task.

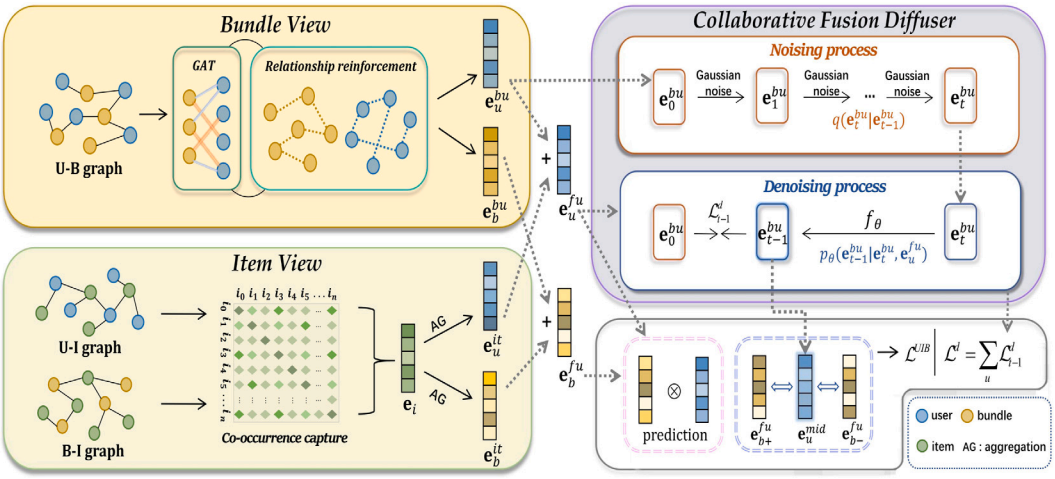


Fig. 2. Overview of the DCDL. To demonstrate our process intuitively, we use a vector as the fundamental unit of representation.

2. Related work

Bundle recommendation. With the rapid development of deep learning technologies, recent methods have made significant progress in optimizing feature representations, enhancing graph structures, and modeling from multiple views. HIDGN [14] introduced modeling the user intent from three perspectives (user-level, bundle-level, and item-level), aiming to capture the diversity of the user intent through cross-task intent contrastive learning from a hierarchical structure. CLBR [15] designed a heuristic sampler to generate counterfactual graph views for graph-based models and introduced counterfactual data augmentation and constraints to mitigate the impact of data sparsity. HyperMBR [16] encoded entities (users, items, and bundles) of two-view interaction graphs in hyperbolic space and proposed a distillation method based on hyperbolic distance to facilitate knowledge transfer between these two views. GPCL [17] considered each user/bundle/item embedding as a Gaussian distribution and designed a contrastive learning module to capture contextual information while mitigating sampling bias issues. Although many methods perform well, the failure to address the negative effects of view information fusion continues to limit further performance improvements.

Diffusion learning. Diffusion learning was initially applied in the image generation field. The complete process involves gradually adding noise to the data and then learning how to progressively recognize the noise to restore the original data distribution, thereby generating complex data distributions from random noise. NCSN [18] introduced a new generative model where samples are generated using the data distribution gradients estimated through Langevin dynamics and score matching. This process approach perturbs data with Gaussian noise at different levels and jointly estimates the corresponding scores. DDPM [19] refined the diffusion learning process with simple modifications, achieving competitive log-likelihoods while maintaining high-quality sample generation. DiffRec [20] explores dimensionality compression by clustering items and reweighting user interactions based on timestamps to encode temporal information. DreamRec [21] generates an oracle item through the denoising process to recover positive items, eliminating the need for negative sampling and directly capturing users' true preferences. However, diffusion-based methods in sequential recommendation [22] scenarios primarily focus on single interaction types, resulting in generated target items that fail to enhance recommendation diversity in multi-view contexts.

3. Methodology

Let $\mathcal{U}$, $\mathcal{B}$, and $\mathcal{I}$ represent the sets of users, bundles, and items respectively, denoted as $\mathcal{U} = \{u_1, u_2, \ldots, u_{|\mathcal{U}|}\}$, $\mathcal{B} = \{b_1, b_2, \ldots, b_{|\mathcal{B}|}\}$, and $\mathcal{I} = \{i_1, i_2, \ldots, i_{|\mathcal{I}|}\}$, where $|\cdot|$ represents the total number of elements in a set. The U–B, U–I, and B–I interactions can build three matrices, recorded as $\mathbf{X} \in \mathbb{R}^{|\mathcal{U}|\times|\mathcal{B}|}$, $\mathbf{Y} \in \mathbb{R}^{|\mathcal{U}|\times|\mathcal{I}|}$, and $\mathbf{Z} \in \mathbb{R}^{|\mathcal{B}|\times|\mathcal{I}|}$. Here, when $x_{ub} \in \mathbf{X}$, $y_{ui} \in \mathbf{Y}$, or $z_{bi} \in \mathbf{Z}$ correspond to an observed interaction, its value is 1, otherwise 0. The goal of the bundle recommendation task is to leverage the observed interaction information to predict unobserved U–B interactions.

3.1. Overview of the approach

For bundle recommendation, the key is to effectively extract interaction information and optimize the fusion of different interaction views. Based on the analysis in the related work section, our proposed DCDL includes three modules: (1) Bundle View Module: Utilizes a GAT layer to process U–B interactions and constructs an augmented relation matrix to enhance information propagation, thereby obtaining feature representations for users/bundles. (2) Item View Module: Captures item similarity through co-occurrence relationships and updates item feature representations. It then aggregates users/bundles' feature representations from items' based on U–I and B–I interactions. (3) Collaborative Fusion Diffuser (CFD): Introduces Gaussian noise to user features in the bundle view and incorporates merged user feature representations from both views during the denoising process. The intermediary state of user features obtained through this process helps adjust the feature space distribution, enhancing recommendation diversity and optimizing collaboration between views. An overview of the DCDL is shown in Fig. 2.

3.2. Bundle view module

In the DCDL's bundle view, a GAT is utilized to identify and amplify significant U–B interactions from limited data information. Following the original GAT framework, users and bundles are treated as nodes within an isomorphic graph according to $\mathbf{X}$, with self-loops added to each node. The user and bundle feature matrices, $\mathbf{E}_u$ and $\mathbf{E}_b$, are initialized with values drawn from a normal distribution. To enhance computational efficiency, feature dimension transformation and multi-head attention are omitted in this implementation. The importance of node n to node m, denoted as c_{mn}, is calculated as follows:

$$c_{mn} = \mathbf{h}^{\top}\left[\mathbf{W}\mathbf{e}_m \parallel \mathbf{W}\mathbf{e}_n\right], n \in \mathcal{N}_m, \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{d\times d}$ is a weight matrix, d is feature dimension, $\mathcal{N}_m$ represents the set of neighboring nodes of node m in the graph and

$\mathbf{h}^\top \in \mathbb{R}^{2d}$ denotes a single-layer feedforward neural network. The subsequent process will be consistent with the original GAT:

$$\alpha_{mn} = \frac{\exp\left(\text{LeakyReLU}\left(c_{mn}\right)\right)}{\sum_{j\in\mathcal{N}_m}\exp\left(\text{LeakyReLU}\left(c_{mj}\right)\right)}, \tag{2}$$

$$\mathbf{e}_m^g = \sigma\left(\sum_{n\in\mathcal{N}_m}\alpha_{mn}\mathbf{W}\mathbf{e}_n\right). \tag{3}$$

In this way, we obtain the updated user feature matrix and bundle feature matrix, $\mathbf{E}_u^g$ and $\mathbf{E}_b^g$.

In the described approach, the GAT is deployed on a U–B interaction graph, focusing solely on observable interactions. However, real-world e-commerce contexts often contain latent interaction data, such as user friendships. To enrich the feature representation in the bundle view, we propose a novel paradigm to implement relationship reinforcement. To achieve that, we first construct an augmented relationship matrix $\mathbf{R}$:

$$\mathbf{R} = \begin{bmatrix}\mathbf{X} & [\mathbf{X}\mathbf{X}^\top] \\ [\mathbf{X}^\top\mathbf{X}] & \mathbf{X}^\top\end{bmatrix}. \tag{4}$$

For $\mathbf{X}\mathbf{X}^\top$ and $\mathbf{X}^\top\mathbf{X}$, a larger element value indicates more same interactions between two users or bundles. By introducing a threshold k_1, elements exceeding k_1 are assigned 1, signifying virtual connections, otherwise 0. Next, inspired by the principal component analysis [23], the output of the bundle view can be calculated as:

$$[\mathbf{E}_u^{bu}\|\mathbf{E}_b^{bu}] = \frac{\mathbf{R}\mathbf{R}^\top}{\|\mathbf{R}\mathbf{R}^\top\|_F}[\mathbf{E}_u^g\|\mathbf{E}_b^g], \tag{5}$$

where $\|\cdot\|_F$ represents the Frobenius norm of a matrix.

3.3. Item view module

An important consideration is that users' choices of bundles are significantly influenced by the items within those bundles. Therefore, effectively modeling item representations remains crucial.

In the context of bundle recommendation, the number of interactions related to items is typically substantial. This means that directly employing complex neural network structures to process information under the item view could introduce additional uncertainty and computational overhead. To address this, our strategy optimizes item feature representation by leveraging item co-occurrence relationships, as indicated by U–I and B–I interactions, to capture similarities between items:

$$\mathbf{C}_u = \frac{\mathbf{Y}^\top\mathbf{Y}}{\sum_j(\mathbf{Y}^\top\mathbf{Y})_{ij}}; \mathbf{C}_b = \frac{\mathbf{Z}^\top\mathbf{Z}}{\sum_j(\mathbf{Z}^\top\mathbf{Z})_{ij}}, \tag{6}$$

where $\mathbf{C}$ represents the matrix obtaining co-occurrence relationships. Next, we employ an additive fusion method with residual connectivity properties to update item feature representations:

$$\mathbf{E}_i^{it} = \mathbf{E}_i + \text{ReLU}(\mathbf{C}_u\mathbf{E}_i) + \text{ReLU}(\mathbf{C}_b\mathbf{E}_i), \tag{7}$$

where $\mathbf{E}_i$ is also initialized using values sampled from a normal distribution. Within the item view, our primary goal is to deduce the user feature $\mathbf{e}_u^{it}$ and the bundle feature $\mathbf{e}_b^{it}$ by leveraging the item features so that we make use of the U–I and B–I interaction graphs and adopt an aggregation strategy:

$$\mathbf{e}_u^{it} = \sum_{i\in\mathcal{N}_u}\frac{1}{|\mathcal{N}_u|}\mathbf{e}_i^{it}; \mathbf{e}_b^{it} = \sum_{i\in\mathcal{N}_b}\frac{1}{|\mathcal{N}_b|}\mathbf{e}_i^{it}. \tag{8}$$

By stacking feature vectors, we can obtain $\mathbf{E}_u^{it}$ or $\mathbf{E}_b^{it}$ directly.

3.4. Collaborative fusion diffuser

Here, we provide a concise overview of the fundamental principles of diffusion learning, illustrated with an example. The DDPM, a prototypical diffusion model, constructs dual Markov chains. It utilizes the probabilistic chain rule in conjunction with the Markov property to approximate the underlying data distribution effectively. The primary objective of the diffusion model is to reconstruct the original data distribution, symbolized as $p_\theta\left(\mathbf{x}_0\right)$, where $\mathbf{x}_0$ signifies the initial state and θ denotes associated parameters. In the forward process, Gaussian noise is gradually added to $\mathbf{x}_0$ following a variance sequence $\left[\beta_1, \beta_2, \ldots, \beta_T\right]$, and T represents the maximum value of t:

$$\begin{aligned} q\left(\mathbf{x}_{1:T} \mid \mathbf{x}_0\right) &= \prod_{t=1}^{T} q\left(\mathbf{x}_t \mid \mathbf{x}_{t-1}\right); \\ q\left(\mathbf{x}_t \mid \mathbf{x}_{t-1}\right) &= \mathcal{N}\left(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right), \end{aligned} \tag{9}$$

where $\mathbf{x}_t$ represents the state of the variable at the step t. α_t is defined as $1-\beta_t$, and $\bar{\alpha}_t$ is defined as $\bar{\alpha}_t = \prod_{s=1}^{t}\alpha_s$. During the reverse denoising phase, the model learns to identify the noise added at each step, and attempts to restore the initial distribution:

$$\begin{aligned} p_\theta\left(\mathbf{x}_{0:T}\right) &= p\left(\mathbf{x}_T\right)\prod_{t=1}^{T} p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right); \\ p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right) &= \mathcal{N}\left(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta\left(\mathbf{x}_t, t\right), \Sigma_\theta\left(\mathbf{x}_t, t\right)\right). \end{aligned} \tag{10}$$

The training phase can be viewed as optimizing the usual variational bound on negative log-likelihood, and the loss can be approximately equal to minimizing the KL divergence between $q\left(\mathbf{x}_{0:T}\right)$ and $p_\theta\left(\mathbf{x}_{0:T}\right)$. It mainly depends on:

$$\mathbb{E}_q\left[\sum_{t\geq 1} D_{\text{KL}}\left(q\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right) \parallel p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right)\right)\right]. \tag{11}$$

With the specific reparameterization $\boldsymbol{\mu}_\theta$, the DDPM's optimization objective at the step t, denoted as $\mathcal{L}_{t-1}^d$, can be calculated as the mathematical expectation:

$$\begin{aligned} \mathcal{L}_{t-1}^d \approx \mathbb{E}_{\mathbf{x}_0,\epsilon}\Bigg[&\frac{\beta_t^2}{2\sigma_t^2\alpha_t\left(1-\bar{\alpha}_t\right)} \\ &\times\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\Bigg], \end{aligned} \tag{12}$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ denotes the noise, $\bar{\alpha}_t$ reflects the accumulated proportion of data retention (not covered by noise) at the step t, and is utilized to calibrate the weighting between noise and current data status. The function approximator ϵ_θ is designed to predict ϵ from $\mathbf{x}_t$, and its specific formulation depends on the requirements of the particular tasks [24,25].

Reverting to the bundle recommendation context, prior research [26] has demonstrated that relying solely on U–B interactions does not yield optimal results. Therefore, the integration of information from different views is essential. However, as mentioned earlier, merging item information without proper adjustments can undermine the diversity of bundle recommendations, thus negatively impacting recommendation performance. Therefore, we aim to capture an intermediate state both before and after feature fusion to guide feature evolution. The diffusion learning process of noising and denoising perfectly enables the acquisition of this intermediate state. Additionally, by dynamically

Algorithm 1 Collaborative fusion diffuser's algorithm

Input: $\mathbf{E}_u^{bu}, \mathbf{E}_u^{fu}$
Output: $\mathbf{E}_u^{mid}, \mathcal{L}_{t-1}^d$

```
for e_u^bu in E_u^bu
    e_0 = e_u^bu                                  Sample initial state.
    t ~ Uniform({1, ..., T})                      Sample diffusion step.
    ε ~ N(0, I)                                   Sample Gaussian noise.
    e_t = sqrt(ᾱ_t) e_0 + sqrt(1 − ᾱ_t) ε         Add noise.
    e_u^mid = f_θ(e_t, e_u^fu, t)                 Calculate the intermediate.
end for
return E_u^mid, L_{t−1}^d
```

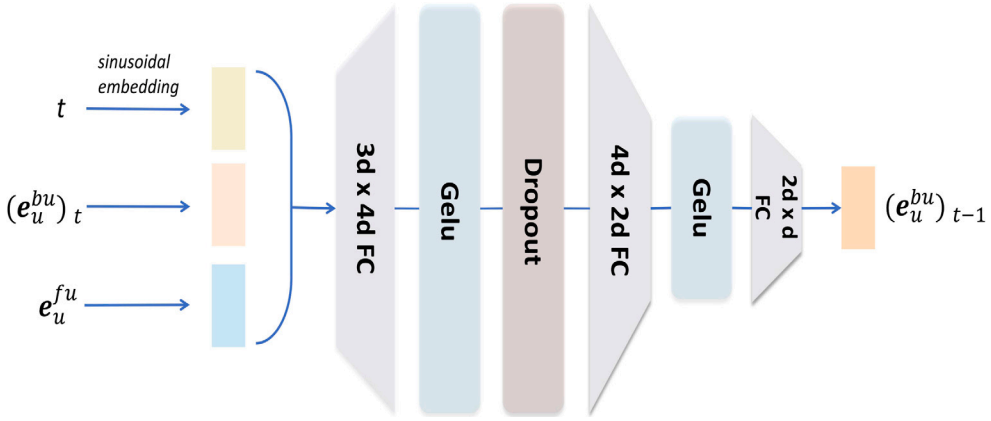


Fig. 3. f_θ's detail structure.

optimizing this intermediary feature state, we seek to achieve a more uniform feature distribution after fusion, thus enhancing the diversity of recommendations. To this end, we first employ an additive fusion approach to facilitate feature integration across views:

$$\mathbf{E}_u^{fu} = \mathbf{E}_u^{bu} + \mathbf{E}_u^{it}; \mathbf{E}_b^{fu} = \mathbf{E}_b^{bu} + \mathbf{E}_b^{it}. \tag{13}$$

Given our goal to predict U–B interactions, we select user features in the bundle view as the initial state and add Gaussian noise:

$$q\left((\mathbf{e}_u^{bu})_t \mid (\mathbf{e}_u^{bu})_{t-1}\right) = \mathcal{N}\left((\mathbf{e}_u^{bu})_t; \sqrt{1-\beta_t}(\mathbf{e}_u^{bu})_{t-1}, \beta_t \mathbf{I}\right). \tag{14}$$

After introducing noise over several steps, we design the function f_θ and its detailed structure, as shown in Fig. 3. The step t's sinusoidal embedding is aligned with DDPM, and the β sequence is sampled using a linearly increasing method, with the value range from 0 to 0.03. Each $\mathbf{e}_u^{bu}$ is assigned a random t within the range of T during every training epoch. f_θ concatenates t's sinusoidal embedding, user features in the bundle view, and the fused user features, aiming to generate the desired intermediary feature state from the denoising process. Additionally, it is necessary to restructure the reparameterization method $\boldsymbol{\mu}_\theta$ to align with our training objective (Detailed derivation proof and explanation can be found in Appendix):

$$\begin{aligned} &p_\theta\left((\mathbf{e}_u^{bu})_{t-1} \mid (\mathbf{e}_u^{bu})_t, \mathbf{e}_u^{fu}\right) = \\ &\mathcal{N}\left((\mathbf{e}_u^{bu})_{t-1}; \boldsymbol{\mu}_\theta\left((\mathbf{e}_u^{bu})_t, (\mathbf{e}_u^{fu}), t\right), \boldsymbol{\Sigma}_\theta\left((\mathbf{e}_u^{bu})_t, (\mathbf{e}_u^{fu}), t\right)\right), \end{aligned} \tag{15}$$

$$\begin{aligned} &\boldsymbol{\mu}_\theta\left((\mathbf{e}_u^{bu})_t, \mathbf{e}_u^{fu}, t\right) = \\ &\sqrt{\bar{\alpha}_t} f_\theta\left((\mathbf{e}_u^{bu})_t, \mathbf{e}_u^{fu}, t\right) + \sqrt{\frac{\alpha_t}{1-\bar{\alpha}_t}}\left(1-\bar{\alpha}_{t-1}\right)\epsilon. \end{aligned} \tag{16}$$

As a result, the training loss $\mathcal{L}_{t-1}^d$ of the CFD at the step t evolves into a newly formulated mathematical expectation expression:

$$\begin{aligned} \mathcal{L}_{t-1}^d \approx \mathbb{E}_{(\mathbf{e}_u^{bu}),\epsilon}\Bigg[\frac{\bar{\alpha}_{t-1}}{2\tilde{\beta}_t}\Bigg\|\mathbf{e}_u^{bu} - \sqrt{\alpha_t} f_\theta\Big(\sqrt{\bar{\alpha}_t}\mathbf{e}_u^{bu} \\ + \sqrt{1-\bar{\alpha}_t}\epsilon, \mathbf{e}_u^{fu}, t\Big)\Bigg\|^2\Bigg]. \end{aligned} \tag{17}$$

Algorithm 1 illustrates the detailed computational steps involved in the CFD for a user in one training epoch. We employ the fused user and bundle features to calculate predictive scores for the users and utilize the UIB [27] loss calculation paradigm to support the intermediary feature state obtained from CFD to guide feature evolution:

$$\hat{y}_{\text{ub}} = \mathbf{e}_u^{fu}(\mathbf{e}_b^{fu})^\top, \mathrm{s} = \mathbf{e}_u^{mid}\mathbf{w}_u^\top, \tag{18}$$

$$\mathcal{L}^{UIB} = \sum_{(u,b^+)} \phi\left(\mathrm{s} - \hat{y}_{ub^+}\right) + \alpha_1 \sum_{(u,b^-)} \phi\left(\hat{y}_{ub^-} - \mathrm{s}\right), \tag{19}$$

$$\mathcal{L} = \mathcal{L}^{UIB} + \lambda_1 \mathcal{L}^d + \lambda_2 \left\|\Theta_2^2\right\|, \tag{20}$$

Table 1
Dataset statistics.

Datasets	#U	#I	#B	#U–I	#U–B	Avg.#I/B
Youshu	8,039	32,770	4,771	138,515	51,377	37.03
NetEase	18,528	123,628	22,864	1,128,065	302,303	77.80
iFashion	53,897	42,563	27,694	2,290,645	1,679,708	3.86

where λ_1, λ_2, α_1 are the hyperparameters, ϕ is LnSigmoid [28], $\mathbf{w}_u \in \mathbb{R}^d$ is a learnable parameter vector, $\left\|\Theta_2^2\right\|$ represents the regularization term, $\hat{\mathbf{y}}_{ub^+}$ represents the score for observed interactions, while $\hat{\mathbf{y}}_{ub^-}$ is the score for non-existent interactions, treated as negative samples. $\mathcal{L}^d$ is the cumulative sum over $\mathcal{L}_{t-1}^d$, and its magnitude needs to be adjusted to align with that of $\mathcal{L}^{UIB}$.

4. Experiment

4.1. Experimental setup

Datasets. The datasets used are Youshu, NetEase, and iFashion, with their specific statistics shown in Table 1. Every dataset is divided into training, validation, and test sets in a ratio of 8:1:1, consistent with the division used in CrossCBR.

Baselines. DCDL is evaluated in comparison with leading SOTA methods, encompassing machine learning approaches such as MFBPR [4] and DAM [5], graph-based models including SGL [29], BundleNet [30], and BGCN [7], and graph-based models employing innovative paradigms, including MIDGN [8], CrossCBR [9] and GPCL [17].

Parameter Settings. For each observed U–B interaction, we randomly select one negative sample and set the initial learning rate from [1e−2, 5e−3, 1e−3, 5e−4]. Entity features are initialized using the He initialization method, with a fixed dimension $d = 64$. The batch size is configured to 2048. We update model parameters using the AdamW optimizer. Both GAT and other neural network layers use a dropout rate of 0.2 to enhance the model's robustness. Hyperparameter k_1 is selected from [2, 4, 6, 8, 10], the diffusion step range T is searched within [5, 10, 15, 20, 25], α_1 is chosen from [4, 8, 12, 16, 20], λ_1 is considered from [0.10, 0.15, 0.20, 0.25, 0.30], and L2 regularization weight λ_2 is tuned from [1e−1, 1e−2, 1e−3, 1e−4]. All models are trained on an RTX3090 GPU using PyTorch 2.0.0.

Evaluation. Following the procedure of GPCL, we employ the Recall and NDCG metrics to evaluate the performance on the three datasets, selecting 20 and 40 as the cut-off ranks. Furthermore, to ensure the reliability and consistency of our results, we conducted each experiment five times on every dataset. The average performance across these five iterations is calculated and reported. The code is available at https://github.com/TH000-Duncan/DCDL.

Table 2
Comparison of different methods.

Methods \ Datasets	Youshu				NetEase				iFashion			
	R@20	*N@20*	*R@40*	*N@40*	*R@20*	*N@20*	*R@40*	*N@40*	*R@20*	*N@20*	*R@40*	*N@40*
MFBPR	0.1959	0.1117	0.2735	0.1320	0.0355	0.0181	0.0600	0.0246	0.0752	0.0542	0.1162	0.0687
DAM	0.2082	0.1198	0.2890	0.1418	0.0411	0.0210	0.0690	0.0281	0.0629	0.0450	0.0995	0.0579
SGL	0.2568	0.1527	0.3537	0.1790	0.0687	0.0368	0.1058	0.0467	0.0933	0.0690	0.1389	0.0851
BundleNet	0.1895	0.1125	0.2675	0.1335	0.0391	0.0201	0.0661	0.0271	0.0626	0.0447	0.0986	0.0574
BGCN	0.2347	0.1345	0.3248	0.1593	0.0491	0.0258	0.0829	0.0346	0.0733	0.0531	0.1128	0.0671
MIDGN	0.2645	0.1530	0.3701	0.1812	0.0669	0.0351	0.1064	0.0435	0.0907	0.0662	0.1327	0.0819
CrossCBR	0.2813	0.1668	0.3785	0.1938	0.0842	0.0457	0.1264	0.0569	0.1173	0.0895	0.1699	0.1080
GPCL	0.2882	0.1713	0.3963	0.2007	0.0833	0.0441	0.1270	0.0557	0.1216	0.0928	0.1756	0.1118
DCDL	**0.3035**	**0.1782**	**0.4032**	**0.2034**	**0.0883**	**0.0473**	**0.1320**	**0.0598**	**0.1324**	**0.1029**	**0.1889**	**0.1231**
Improv. (%)	5.31%	4.03%	1.74%	1.35%	4.87%	3.50%	3.94%	5.10%	8.88%	10.90%	7.57%	10.12%

The underlined value represents the maximum baseline value, and the bold value represents the overall maximum value.

Table 3
Ablation experiment results.

Methods \ Datasets	Youshu		NetEase		iFashion	
	R@20	*N@20*	*R@20*	*N@20*	*R@20*	*N@20*
DCDL_{gcn}	0.2917	0.1725	0.0837	0.0443	0.1258	0.0967
$\text{DCDL}_{/re}$	0.2923	0.1728	0.0828	0.0427	0.1269	0.0965
$\text{DCDL}_{/co}$	0.2954	0.1741	0.0816	0.0421	0.1226	0.0931
$\text{DCDL}_{/cfd}$	0.2833	0.1686	0.0807	0.0414	0.1156	0.0872
DCDL	**0.3035**	**0.1782**	**0.0883**	**0.0473**	**0.1324**	**0.1029**

4.2. Overall performance

Initially, we conduct a comparative analysis of the DCDL's ranking performance against various baselines. Table 2 presents the results on the three datasets. The observations from the table reveal that:

- In all cases, DCDL outperforms baseline models, demonstrating noticeable improvements. This validates the effectiveness of our approach and aligns with our theoretical analysis, confirming the necessity of optimizing the feature distribution after view fusion.
- The findings reveal that methods rooted in traditional machine learning fall short compared to those utilizing graph neural networks in terms of modeling efficacy. Besides, incorporating a new paradigm into the graph-based neural network approach to address specific issues could substantially enhance the model's performance.
- The experimental results also show that DCDL has a significant advantage on the iFashion dataset, but its lead is less pronounced on the Youshu dataset, particularly at a cut-off rank of 40, where its performance nearly levels with GPCL. This is in line with our expectations, as according to γ analysis, the U–B interaction in the Youshu dataset relies less on item information, so it does not seriously damage the diversity of bundle recommendations during the fusion process, while iFashion does the opposite.

4.3. Ablation study

In the design of the ablation study, we have created four sets of experiments with distinct structures and symbolic representations to thoroughly assess the effectiveness of the components of DCDL. These are as follows:

- DCDL_{gcn}: Replaces the GAT in DCDL with a graph convolutional network.
- $\text{DCDL}_{/re}$: Removes the relationship reinforcement in the bundle view.
- $\text{DCDL}_{/co}$: Removes the co-occurrence capture in the item view.
- $\text{DCDL}_{/cfd}$: Removes the CFD (Directly use $\mathbf{e}_u^{fu}$ and $\mathbf{e}_b^{fu}$ to predict based on Bayesian Personalized Ranking loss [4].

Table 3 demonstrates that various components of DCDL contribute to the bundle recommendation task to different extents. In the NetEase and iFashion datasets, the model performs well even without the GAT or the relation reinforcement method. This suggests that attention mechanisms may face challenges in converging as the amount of U–B interaction information increases. The co-occurrence capture also demonstrates a certain level of effectiveness. Additionally, the CFD significantly affects model performance, surpassing other components. The absence of CFD notably affects the NetEase and iFashion datasets, aligning with prior γ analyses. U–B interactions in Youshu inherently show rich diversity and are less reliant on item information, meaning that integrating item information does not overly disrupt them. Consequently, CFD offers greater potential for enhancement in the NetEase and iFashion datasets.

4.4. Hyperparameter studies

We selected four key hyperparameters for experimentation, and the results are depicted in Fig. 4. Priority was given to Recall@20 as the evaluation metric. This section will analyze each hyperparameter individually.

4.4.1. The hyperparameter k_1

k_1 serves as the threshold for establishing virtual connections in the U–B interactions. The results indicate a slight peak at $k_1 = 4$ for the Youshu dataset, but its impact is irregular in the other two datasets. Overall, the choice of k_1 has a minor effect on the model's performance across all three datasets. This can be attributed to the dual-view model structure, which provides more tolerance to fluctuations in hyperparameter values within individual views.

4.4.2. The hyperparameter T

T denotes the step range in the noise-adding process, with a higher T enabling greater perturbation of initial features. Experimental results suggest that the optimal value of T increases with the richness of U–B information in the dataset, aligning with the overall performance from another perspective. Furthermore, the observed 'quick rise and slow fall' trend in the Youshu and NetEase datasets corroborates the effectiveness of the noising algorithm, which progressively adds noise in a balanced manner by considering the cumulative degree of perturbation.

4.4.3. The hyperparameter λ_1

λ_1 determines the degree of emphasis placed on CFD's restoration ability during training. The line graphs for all three datasets show a peak as the value of λ_1 increases, with iFashion's occurring later. We infer that this behavior is characteristic of datasets with rich U–B

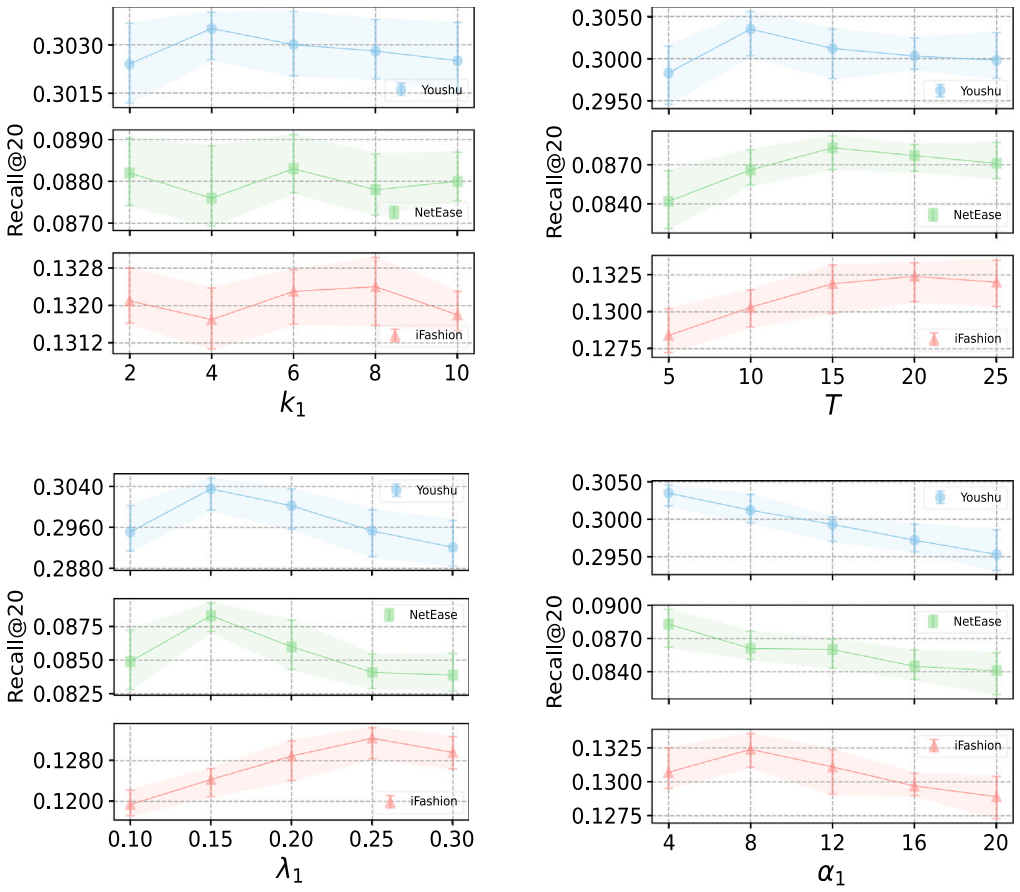


Fig. 4. Hyperparameter experiment results (the shaded area represents the fluctuation range of values from multiple experiments).

information, where user features under the bundle view are less stable in the early stages of training, making the restoration process more challenging. This indicates that the DCDL is relatively sensitive to λ_1, highlighting the crucial auxiliary role of the CFD in guiding feature evolution. An unbalanced weight can lead to a misplaced emphasis on training because our design did not force CFD to strictly revert to the initial feature state.

4.4.4. The hyperparameter α_1

α_1 is used to balance the contributions of positive and negative samples. Experimental results show that across all three datasets, an increase in α_1 leads to a moderate decrease in performance. This indicates the robustness of our design and suggests that our approach does not rely heavily on sharply distinguishing between positive and negative sample pairs for guiding feature changes. Instead, it aligns with our objective of bringing mild random interference, rather than imposing strong constraints on the process.

4.5. Further analysis

4.5.1. Time complexity

In DCDL we only embed the newly designed training phase paradigm of diffusion learning, and exclude its generation phase, which means that compared with CrossCBR or other dual-view modeling methods, our time complexity still grows linearly. Considering the main matrix multiplication and diffusion learning parts, the time complexity for an epoch can be presented as $O((|\mathcal{U}|+|\mathcal{B}|)d^2+(|\mathcal{I}|^2+(|\mathcal{U}|+|\mathcal{B}|)^2)d+|\mathcal{U}|(T+C)d)$ approximately, where d represents feature dimension (64 in our work), T represents the average diffusion step, and C signifies the time overhead associated with the denoising phase (primarily the computational cost on f_θ).

4.5.2. CFD discussion

In CFD, our core idea is to use the diffusion learning paradigm to bring controllable randomness into the fused user feature representation. This approach helps explore a broader feature space, prevents excessive feature clustering, and ultimately enhances the diversity of recommendations.

In the process, we also consider whether simpler network structure, noise methods, or generative methods [20,31–34] can provide similar effects. For this reason, we selected 2 layers MLP [35] and VAE [36] to conduct replacement experiments on CFD respectively. We also consider adding random Gaussian noise directly on $\mathbf{e}_u^{fu}$ without $\mathbf{e}_u^{mid}$(RN) for comparison. Regarding MLP and VAE, we concatenate the user feature vector in the bundle view and the fused user feature vector as input, and the output is still regarded as $\mathbf{e}_u^{mid}$. The Recall@20 and NDCG@20 results of the replacement experiment are shown in Fig. 5(a) and Fig. 5(b), respectively. The experimental results are in line with our expectations: irregular and aggressive scaling from neural networks is not well-suited for the recommendation scenario where view collaboration relies solely on interactive data. In such cases, it is difficult to identify a reliable basis for the entire generation process, so moderate disruption of the original features proves to be a more effective choice.

To further validate that CFD operates according to our approach, we visualized the spatial distribution of user features after training for iFashion dataset. Fig. 6(a) shows the feature distribution after training with DCDL$_{/cfd}$ structure, while Fig. 6(b) shows the feature distribution after training with complete DCDL. This visualization with a density curve clearly shows that with CFD processing, the overall feature distribution becomes more dispersed, which enhances the diversity of bundle recommendations. Besides, as shown in Fig. 7, we visualized the recommendation results for some users. Compared to CrossCBR, DCDL

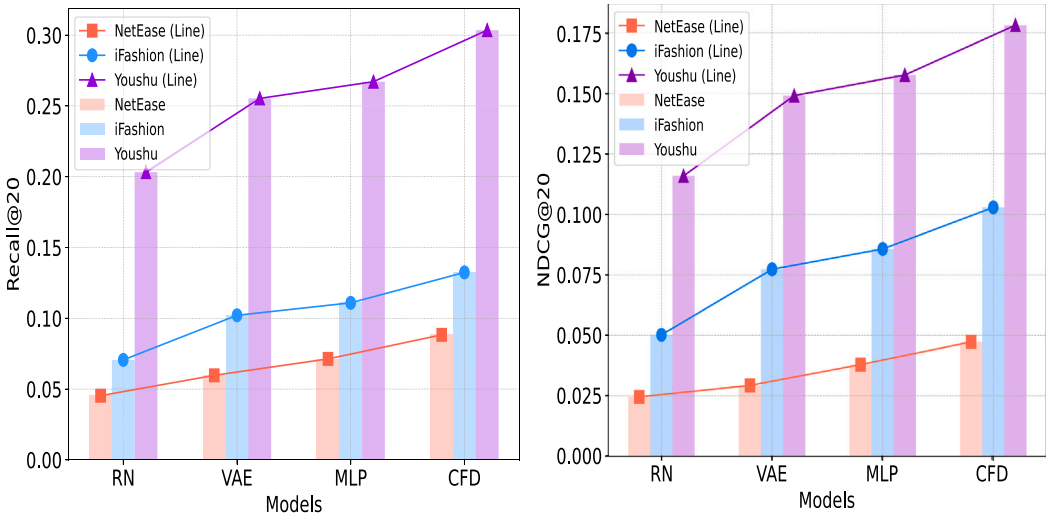


Fig. 5. CFD replacement experiment.

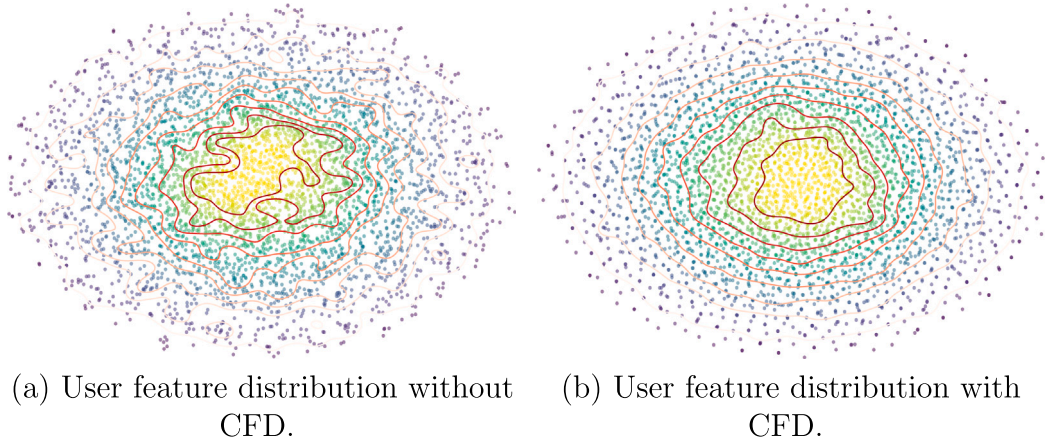


Fig. 6. Visualization of user feature distribution obtained through the t-SNE method.

better captures a broader range of diverse user choices (highlighted by red bundle IDs) in the test set.

The underlying diffusion learning paradigm and collaborative fusion mechanism have considerable potential for broader applicability across various recommendation tasks. One promising direction is to extend CFD into multi-view information fusion scenarios, where its collaborative diffusion mechanism can effectively integrate heterogeneous data from diverse sources or modalities. Specifically, CFD can iteratively refine user or item representations by denoising intermediate states guided by multiple data views – such as user interactions, textual content, image data, and social relationships – thus effectively aligning and harmonizing these disparate sources of information. From the perspective of multimodal information processing, by incorporating textual descriptions, images, or user reviews alongside traditional interaction data, the diffusion learning process could facilitate smoother transitions across different modalities. Such integration is expected to significantly improve both the recommendation quality and interpretability, enabling a richer and more comprehensive user modeling experience. Additionally, the CFD framework naturally lends itself to multi-task learning environments, where its diffusion-based fusion process can generate balanced, generalized representations that support several related recommendation tasks simultaneously (e.g., predicting clicks, likes, and purchase behaviors). This multi-task capability not only enhances representation robustness and transferability but also reduces redundancy, improving the overall performance across tasks. Moreover, CFD can be adapted for session-based recommendation tasks by fusing session-specific contextual data with historical user behaviors. By generating intermediate representations that maintain an optimal balance between recency and diversity, CFD could effectively mitigate challenges like session drift and enhance recommendation quality in real-time applications.

Nevertheless, the practical deployment of the proposed DCDL framework also faces several inherent limitations that merit further consideration. While the diffusion learning paradigm significantly enhances diversity, it inherently introduces additional computational complexity during training, particularly due to iterative denoising processes. While our current approach minimizes overhead by employing a single-step diffusion per training epoch, extremely large-scale datasets (e.g., tens or hundreds of millions of users or items) might pose challenges related to memory consumption and computational cost. Addressing this limitation could involve developing more efficient diffusion training methods, such as sampling-based or hierarchical diffusion techniques that selectively apply the diffusion process to critical subsets of users or items. Future work should explore simplified inference strategies or approximations of the diffusion process that retain recommendation quality while significantly reducing runtime overhead. Finally, deploying the CFD in practical environments might necessitate incremental or online model updating mechanisms, which need exploring adaptive

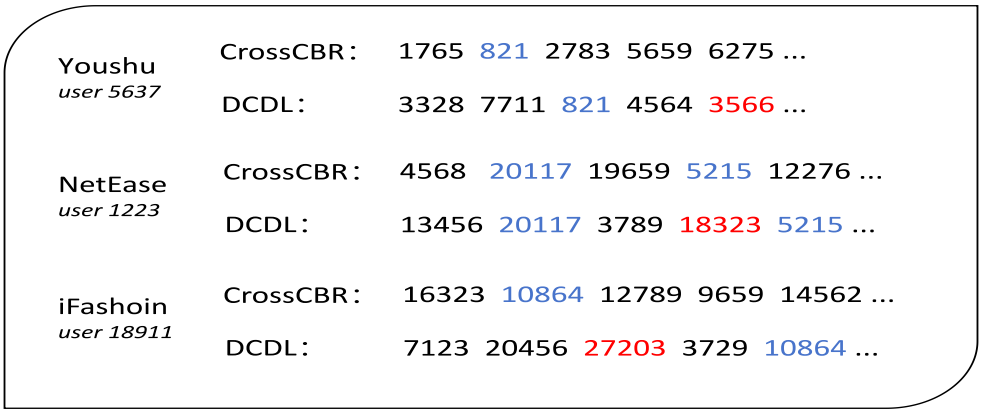


Fig. 7. Colored IDs represent the bundles the user has interacted with: blue indicates the bundle contains items the user has interacted with, while red signifies it does not.

diffusion approaches that dynamically adjust model parameters based on streaming data could enhance applicability in rapidly changing real-world recommendation settings.

5. Conclusion

This paper introduces a novel framework, DCDL, which leverages diffusion learning to mitigate the negative impact of reduced diversity in bundle recommendations during the view information fusion process. Specifically, we first construct a dual-view model that accounts for the different interaction types. Furthermore, we design a diffusion learning module that utilizes a specially parameterized function. This module ensures a more uniform user feature distribution after fusion, thereby enhancing recommendation diversity and improving view collaboration. Experiments conducted on three benchmark datasets demonstrate the exceptional performance and robustness of DCDL. In future research, we aim to further develop a unified CFD application solution, applicable beyond the realm of bundle recommendation.

CRediT authorship contribution statement

Jinpeng Chen: Writing – original draft, Methodology, Conceptualization. **Jinqing Wang:** Writing – original draft, Methodology, Data curation, Validation, Investigation. **Fan Zhang:** Investigation, Formal analysis. **Yuan Cao:** Validation, Investigation. **Kaimin Wei:** Writing – review & editing, Validation. **Liwei Huang:** Writing – review & editing. **Haoran Zhu:** Software. **Feifei Kou:** Investigation, Funding acquisition. **Hongben Huang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Derivation of the rationality of the μ_θ in DCDL

The generation task in Denoising Diffusion Probabilistic Models (DDPM) can be viewed as optimizing a variational bound on the negative log-likelihood. Within its framework, the term corresponding to the tth step is articulated as follows:

$$\mathcal{L}_{t-1} = D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)). \tag{A.1}$$

Since $q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)$ and $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ are both Gaussian distributions, instead of relying on high-variance Monte Carlo estimates, can be computed in a Rao-Blackwellized manner, enabling closed-form solutions:

$$\mathcal{L}_{t-1} = \mathbb{E}_q\left[\frac{1}{2\tilde{\beta}_t}\left\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t,t)\right\|^2\right] + C, \tag{A.2}$$

where C is a constant and $\tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t,\mathbf{x}_0\right)$ can be calculated as:

$$\tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t,\mathbf{x}_0\right) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathbf{x}_t. \tag{A.3}$$

For $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$, we further get:

$$\begin{aligned}\tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t,\mathbf{x}_0\right) &= \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathbf{x}_t \\ &= \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\epsilon}{\sqrt{\bar{\alpha}_t}} + \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathbf{x}_t \\ &= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right).\end{aligned} \tag{A.4}$$

In DDPM, $\boldsymbol{\mu}_\theta$ is reparametrized as:

$$\boldsymbol{\mu}_\theta\left(\mathbf{x}_t,t\right) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta\left(\mathbf{x}_t,t\right)\right). \tag{A.5}$$

According to Eq. (A.2), (A.4) and (A.5), we get the simplified version of $\mathcal{L}_{t-1}$:

$$\mathcal{L}_{t-1} = \mathbb{E}_{\mathbf{x}^0,\epsilon}\left[\frac{\beta_t^2}{2\tilde{\beta}_t\alpha_t\left(1-\bar{\alpha}_t\right)}\left\|\epsilon - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] + C. \tag{A.6}$$

It is necessary to restructure $\boldsymbol{\mu}_\theta$ to align it with our specific scenario requirements. Therefore, we propose a new $\boldsymbol{\mu}_\theta$ form (Note that, in this

context, we do not incorporate f_θ with auxiliary information, as it does not affect the overall structure of the formula):

$$\boldsymbol{\mu}_\theta\left(\mathbf{x}_t, t\right) = \sqrt{\bar{\alpha}_t} f_\theta\left(\mathbf{x}_t, t\right) + \sqrt{\frac{\alpha_t}{1-\bar{\alpha}_t}}\left(1-\bar{\alpha}_{t-1}\right)\epsilon. \tag{A.7}$$

Then we can derive:

$$\begin{aligned}
&\tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t, \mathbf{x}_0\right) - \boldsymbol{\mu}_\theta\left(\mathbf{x}_t, t\right) \\
=&\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right) - \left(\sqrt{\bar{\alpha}_t} f_\theta\left(\mathbf{x}_t, t\right) + \sqrt{\frac{\alpha_t}{1-\bar{\alpha}_t}}\left(1-\bar{\alpha}_{t-1}\right)\epsilon\right) \\
=&\frac{1}{\sqrt{\alpha_t}}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right) - \sqrt{\bar{\alpha}_t} f_\theta\left(\mathbf{x}_t, t\right) - \sqrt{\frac{\alpha_t}{1-\bar{\alpha}_t}}\left(1-\bar{\alpha}_{t-1}\right)\epsilon \\
=&\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{\frac{\alpha_t}{1-\bar{\alpha}_t}}\left(1-\bar{\alpha}_{t-1}\right)\epsilon - \sqrt{\bar{\alpha}_t} f_\theta\left(\mathbf{x}_t, t\right) - \sqrt{\frac{\alpha_t}{1-\bar{\alpha}_t}}\left(1-\bar{\alpha}_{t-1}\right)\epsilon \\
=&\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 - \sqrt{\bar{\alpha}_t} f_\theta\left(\mathbf{x}_t, t\right).
\end{aligned} \tag{A.8}$$

Thus, we have successfully eliminated the noise-related parameter terms, refocusing the task's objective. As a result, $\mathcal{L}_{t-1}$ simplifies to the following new form:

$$\mathcal{L}_{t-1} = \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[\frac{\bar{\alpha}_{t-1}}{2\tilde{\beta}_t}\left\|\mathbf{x}_0 - \sqrt{\alpha_t} f_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] + C. \tag{A.9}$$

Consequently, we have transformed the loss calculation version in the DDPM equation, which originally focused on predicting noise, into a version that predicts the original user features, while maintaining theoretical equivalence. Our objective is to use the intermediary feature state $\mathbf{e}_u^{mid}$ (the product of the denoising phase) to guide feature changes, without the necessity of reconstructing the original user features from pure noise. Therefore, we introduced the $\sqrt{\alpha_t}$ to ensure that the predictive vector does not entirely concentrate on restoring original features, enhancing the generalizability of $\mathbf{e}_u^{mid}$. Furthermore, by utilizing only the training phase of diffusion learning (excluding its generation or sampling phase), we eliminate the need for separately training the diffusion model, saving both time and computational resources.

Data availability

Data will be made available on request.

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